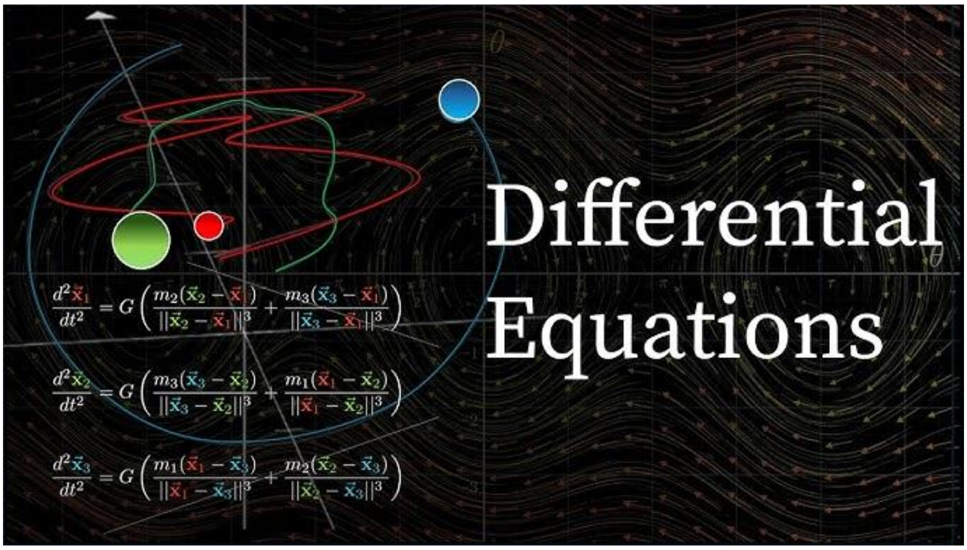
Lect 11: Differential Equation - II

Tuesday, 14 April 2026 11:45 am



Linear Differential Equations

Example

Solve the differential equation

$$y' - 3y = 6$$

Example

Solve the differential equation

$$y' - 2xy = x$$

Solution:

$$y = \frac{-1}{2} + ce^{x^2}$$

Example

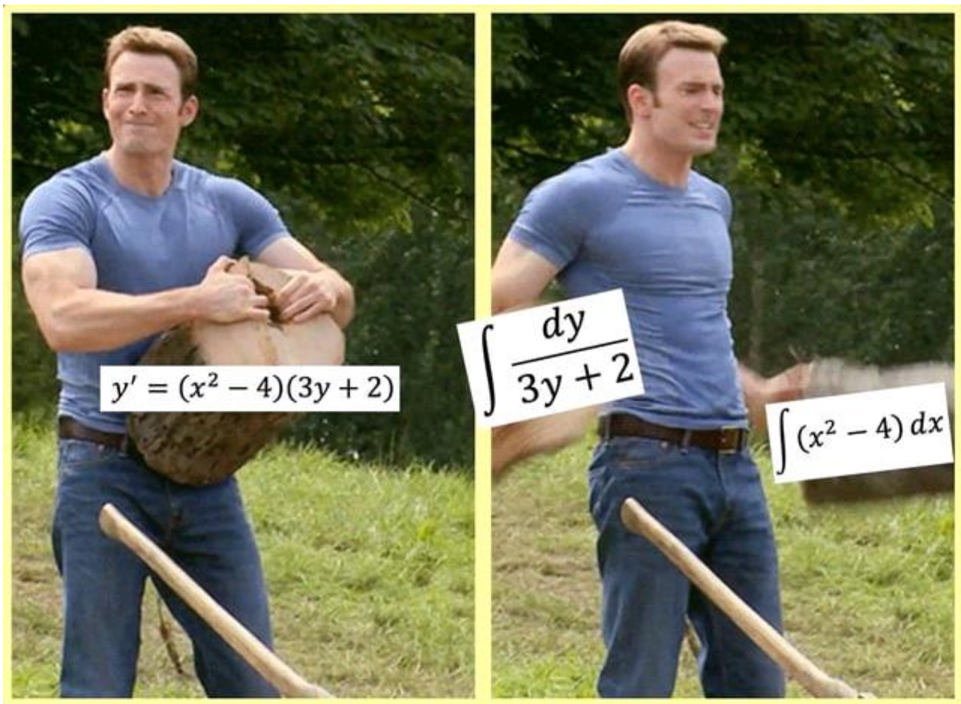
Solve the differential equation

$$y' + y = \sin x$$

$$y = -2 + ce^{3x}$$

$$y = \frac{-1}{2} + ce^{x^2}$$

$$y = \frac{1}{2}(\sin x - \cos x) + ce^{-x}$$



Differential Equation 2: Variable Separable & Integrating Factor

Sunday, 29 October 2023 9:47 pm

→ Variable-Separable ODEs

$\dagger$: $\boxed{\dfrac{dy}{dx} = \boxed{F(x,y)}}$ ← 1^{st} Order ODE

$\dagger$ is variable-separable if $F(x,y) = \phi(x)\psi(y)$

i.e. $\dfrac{dy}{dx} = \boxed{\phi(x) \cdot \psi(y)}$ —— (*)

(*) is a variable-separable ODE.

The separation of variables:

$$\frac{dy}{\psi(y)} = \phi(x)\,dx \quad \text{—— } (\tilde{*})$$

Integrate both sides of $(\tilde{*})$

X-integral

Y-integral → $\boxed{\int \dfrac{dy}{\psi(y)} = \int \phi(x)\,dx + C}$ ← This is the general solution of (*).

If $F(x,y) \neq \phi(x)\,\psi(y)$ then No variable separable possible

Ex#1

$$\frac{dP}{dt} = kP, \quad \text{—— (1)}$$

$$P(t) = Ce^{kt},$$

$$\frac{dP}{P} = k\,dt$$

$$\int \frac{1}{P}\,dP = \int k\,dt + C$$

$$\ln(P) = kt + C$$

$$P = e^{kt + C}$$

$$P = e$$
$$P = e^{kt} e^{c}$$
$$P = C_1 e^{kt}$$

Example 2#

$$\frac{dy}{dx} = y^2 x^2 e^{x} \quad \text{--- (i)}$$

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Separating the varinbles

$$\int \frac{dy}{y^2} = \int x^2 e^{x} dx + C$$

$$\int \frac{dy}{y^2} = -\frac{1}{y}$$

$$\int \underbrace{x^2}_{u} \underbrace{e^{x} dx}_{dv} = uv - \int v\, du$$
$$= x^2 e^{x} - 2\int x e^{x} dx$$

$$= x^2 e^{x} - 2[x e^{x} - e^{x}]$$
$$= x^2 e^{x} - 2x e^{x} + 2e^{x}$$

Our general solution of (i)

$$\boxed{-\frac{1}{y} = x^2 e^{x} - 2x e^{x} + 2e^{x} + C.}$$

Verification: (By means of implicit differentiation)

$$\frac{1}{y^2}\frac{dy}{dx} = \cancel{2x e^{x}} + x^2 e^{x} - \cancel{2e^{x}} - \cancel{2x e^{x}} + \cancel{2e^{x}} + 0$$
$$\frac{1}{y^2}\frac{dy}{dx} = x^2 e^{x}$$

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$$\frac{1}{y^2}\frac{dy}{dx} = x^2 e^{x}$$

Ex#3

$$(x-1)(x+2)\frac{dy}{dx} = \frac{1}{y}$$

Separating the variables:

$$\frac{1}{y^2}\frac{dy}{dx} = x^2 e^x$$

Ex#3

$$(x-1)(x+2)\frac{dy}{dx} = \frac{1}{y}$$

Separating the variables:

$$\int y\,dy = \int \frac{dx}{(x-1)(x+2)} + C.$$

Find A and B:

$$\int y\,dy = y^2/2 \qquad \frac{1}{(x-1)(x+2)} = \frac{A}{x-1} + \frac{B}{x+2}$$

$$\int \frac{dx}{(x-1)(x+2)} = \int \frac{A}{x-1}dx + \int \frac{B}{x+2}dx$$

$$= A\ln(x-1) + B\ln(x+2)$$

Our general solution:

$$\boxed{\frac{y^2}{2} = A\ln(x-1) + B\ln(x+2) + C.}$$

Verification: (Implicit Differentiation)

$$y\frac{dy}{dx} = \frac{A}{x-1} + \frac{B}{x+2} = \frac{1}{(x-1)(x+2)}$$

Ex: $x y' - y = x^2$

Ex: $\frac{\sin^{-1}y}{x^3}\frac{dy}{dx} = 7 \qquad \frac{\sin^{-1}(y)}{x^3}\frac{dy}{dx} = 7$

Ex: $\frac{\sqrt{y}}{x^2 \sin x}\frac{dy}{dx} = y/x.$

Ex: $x\,dy - 2\,dy = x\,dx$

The Standard Linear ODE of First-Order:

$$\frac{dy}{dx} + P(x)y = Q(x)$$

Note: if $Q(x) \equiv 0$, (*) is a variable-separable ODE.

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Separable ODE.

$$\frac{dy}{dx} = -P(x)y$$

$$\Rightarrow \frac{dy}{y} = -P(x)dx$$

$$\ln y = -\int P(x)dx + c$$

$$\therefore y = c\, e^{-\int P(x)dx}$$ The General Solution.

Variable Seperable Case

Back to the General Case:

$$\frac{dy}{dx} + P(x)y = Q(x).$$

Comment: Details later!

$$\mu = \mu(x) \leftarrow \text{Integrating Factor}$$
$$= e^{\int P(x)dx}$$

$$e^{\int P(x)dx} \cdot y = \int e^{\int P(x)dx} Q(x) + c$$

$$y = e^{-\int P(x)dx}\left[\int e^{\int P(x)dx} Q(x) + c\right].$$

OR

$$\mu\, y = \int \mu\, Q(x) + c$$

$$\boxed{y = \frac{1}{\mu(x)}\left(\int \mu(x)\, Q(x) + c\right)}$$

Ex: $\frac{dy}{dx} + \frac{y}{x} = \boxed{e^{x^2}}$ Here $\therefore P(x) = \frac{1}{x}$ & $Q(x) = e^{x^2}$

Formulate Eq. in Standard Order Form $\int P(x)dx$ $e^{\ln x} = x$

* Formulate Eq. in Standard Order ...

* Find $P(x) \to \mu(x) = I.F = e^{\int P(x)dx} = e^{\ln x} = x$

* $y = \frac{1}{\mu(x)}\left(\int \mu(x)\, Q(x)\, dx + c\right) = y = \frac{1}{x}\int x e^{x^2} dx + c$

$P(x) = \frac{1}{x}$; $Q(x) = e^{x^2}$

$\therefore \mu = e^{\int p(x)dx} = e^{\int \frac{dx}{x}} = e^{\ln x} = x$

$\therefore xy = \int x e^{x^2} dx + c$

Let $u = x^2 \Rightarrow du = 2x\,dx$

$= \frac{1}{2} e^{x^2} + c$

$\frac{1}{2}\int e^u \underbrace{2x\,dx}_{du}$

$\therefore y = \frac{1}{2x} e^{x^2} + \frac{c}{x}$. ← The General Solution.

$\frac{1}{2}\int e^u du$

Example 4 Find the solution to the following IVP.

$$t y' + 2y = t^2 - t + 1 \quad y(1) = \frac{1}{2}$$

* St. Form $y' + \underbrace{\frac{2}{t}}_{P(x)} y = \underbrace{t - 1 + \frac{1}{t}}_{Q(x)}$

* Integrating Factor $(I.F) = \mu(x) = e^{\int P(x)\,dx}$

$\mu(x) = e^{\int \frac{2}{t} dt} = t^2$

* $y = \frac{1}{\mu(x)}\left(\int \mu(x)\, Q(x)\, dx + c\right)$

$$y = \frac{1}{t^2}\left(\int t^2\left(t - 1 + \frac{1}{t}\right) dt + C\right)$$

$$y(t) = \frac{1}{t^2}\int (t^3 - t^2 + t)\, dt + C$$

$$= \frac{1}{t^2}\left(\frac{t^4}{4} - \frac{t^3}{3} + \frac{t^2}{2}\right) + \frac{1}{t^2}C$$

$$y(t) = \frac{t^2}{4} - \frac{t}{3} + \frac{1}{2} + \frac{C}{t^2}$$

Substitute $y(1) = \frac{1}{2}$

$$y(1) = \frac{1}{4} - \frac{1}{3} + \frac{1}{2} + \frac{C}{1}$$

$$\frac{1}{2} = -\frac{1}{12} + \frac{1}{2} + C$$

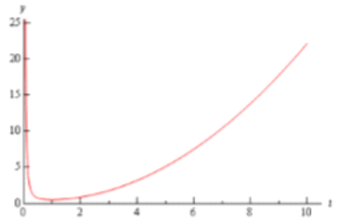
$$C = \frac{1}{12}$$

$$y(t) = \frac{t^2}{4} - \frac{t}{3} + \frac{1}{2} + \frac{1}{12t^2}$$

The solution is then,

$$y(t) = \frac{1}{4}t^2 - \frac{1}{3}t + \frac{1}{2} + \frac{1}{12t^2}$$

Here is a plot of the solution.



Linaer Differential Equation

To solve the linear equation

$$\frac{dy}{dx} + P(x)y = Q(x)$$

we multiply by the integrating factor

$$\mu(x) = e^{\int P(x)\,dx}$$

Then we get

$$[y\mu(x)]' = \mu(x)Q(x)$$

$$y = e^{-\int P(x)\,dx}\left[\int \mu(x)Q(x)\,dx + c\right]$$